

State RegData 2023 Technical Documentation

QuantGov

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1 Purpose

State RegData leverages the power of the QuantGov platform to seamlessly acquire and analyze state regulations and statutes, transforming extensive pages of intricate legal text into valuable datasets. These datasets encompass a wealth of information that includes highly-structured metadata, regulatory restriction counts, complexity metrics, and economic industry relevance. RegData 2023 introduces newly enhanced "document reference" metadata, enabling users to track specific regulatory and statutory documents over time. This user guide will cover RegData 2023 fully, including new features, methodology, and how to access the data in its various forms.

2 Content

As of October 27, 2023, there is data for 48 states and the District of Columbia in the State RegData project, which includes both state regulatory and statutory codes.¹ The datasets include a variety of data including meta-data about the documents and metrics that quantify the text in a variety of ways.

¹We are missing regulatory data for Arkansas and West Virginia. Arkansas does not have a codified regulatory code and West Virginia's regulatory website has data quality issues. We do have statutory data for both states.

2.1 Metadata

The metadata about each document enables user to identify the document and gives some important information about the document.² The metadata for State RegData 2023 includes a variety of columns: "document reference," "document name," "agency," and "cluster." See [Appendix A](#) for explanations of each column.

2.1.1 New Features

State RegData 2023 introduces cleaner "document reference" and "document name" data that more easily allows users to compare specific documents across time. This data can be used to track the changes of specific regulations or statutes over the time span of the dataset. This improved data can be found in the bulk download or by accessing document-level data through [RegHub](#) or directly through the API (see below).

Similar to last year's release, this release includes further improved agency and "cluster" data for state regulations and "cluster" data for state statutes. The agency data refers to the agency responsible for writing or enforcing the regulation, while the clusters attempt to group the agencies into categories. Clusters enable the user to better compare topic-specific regulations or statutes between states, in a way that is more difficult to do with only agency-level data.³

2.2 Data Series

The state datasets include a count of the total number of words and restrictions in each part of the state's regulatory or statutory code, as well as complexity metrics and industry⁴ data. See [Appendix B](#) for explanations of

²The metadata formatting between the states is inconsistent since every state has a different way of organizing its regulatory code. Michigan's code, for example, has a single level of hierarchy, while Washington breaks its code down by title and chapter.

³The "cluster" data in this State RegData version was generated by manually categorizing agencies into clusters. Extensive care and effort were invested to ensure the accuracy of this categorization, but occasional judgment calls were made when assigning agencies to clusters in rare cases.

⁴Each state-level industry dataset includes the number of industry relevant restrictions pertaining to the 3-digit NAICS industry classification for each document, which is calculated by multiplying the probability that the regulatory code part in question pertains to the NAICS industry classification in question by the number of restrictions in that part.

each data series.

Some improved complexity metrics were included in this release, including an improved "conditionals" and "long word score." See the [RegData U.S. 5.0 User's Guide](#) for more information about the improved complexity metrics.

3 Accessing the Data

3.1 Exploring and Visualizing the Data

For users that simply want to explore the data we have already made, we created several tools on quantgov.org that allow users to visualize the data created from the State RegData project.

The first tool is the [State RegData Dashboard](#). The dashboard allows the user to compare restriction counts across every state for the year 2023. By clicking into each state, one can see more data containing our analysis of the state's regulatory and statutory codes, including word and restriction counts, top regulated industries, and agency-level data.

A second tool is the [RegData Explorer](#) interactive. This tool provides a map to explore the state-level data and provides comparisons, rankings, and other useful information.

Another tool that allows users to explore our data and document is RegHub. Using the search tool at <https://reghub.ai/search> (requires free login), users can search for keywords in all documents in the QuantGov corpus, with the ability to filter by jurisdiction, year, topic, and document type (i.e. regulations or statutes). The full text of the search results can then be downloaded by supplying an email address.

3.2 Downloading the Data

In addition to exploring the data using the QuantGov tools, there are several avenues for downloading the data to explore or analyze for yourself. These tools include bulk downloads, an interactive downloader (for both data and documents), and an API (either direct calls or through a Python library). Bulk downloads can be found at <https://reghub.ai/data/bulk>. The downloads are organized by data project. The datasets for State RegData 2023 can be found close to the top, just under the download for RegData U.S. Federal 4.1. The downloads are split into regulations and statutes to minimize

the mixing of the two data sources. Each download contains document-level data, including word and restriction counts along with agency and cluster data. A separate dataset contains industry data which can be merged into the main dataset using the "document id" column. These document-level bulk downloads are intended for researchers looking to explore or analyze our data in detail.

The custom data downloader can be found at <https://reghub.ai/data/custom>. This tool allows the user to download both summary and document-level data for State RegData and other data projects. The data downloader allows the user to select one or more jurisdictions, document type (i.e. regulations or statutes), summary or document level data, the series (i.e. restriction counts, word counts, industry probabilities, etc.), and years. The interactive downloader is intended to provide smaller, more specific datasets than the bulk downloads, without requiring the user to have knowledge about Python or the API.

The Python library for the API can be downloaded using `pip install regcensus` or directly from GitHub. See the README on the GitHub page for more details about how to use the package, or watch the video tutorial at <https://www.quantgov.org/quantgov-api>. The Python library is intended for users with experience using Python who would like to access more specific data not available through our bulk downloads or interactive downloader tool.

4 Technical Notes

- In an attempt to rectify the large differences in which states publish regulatory codes, the State RegData project attempted to aggregate each state's code to a unit that is roughly similar. For example, while California organizes their code all the way to the rule level, we aggregated at the chapter level. This allows for easier comparison in both the metadata and the industry relevance results.
- In approximately half of the states, we sourced text from the [Case-text](#) repository rather than the official website. Casetext often offered cleaner, more consistent text and metadata, as well as a smoother web scraping experience. If errors or inaccuracies were identified in the Casetext repository, we reverted to the official government website as

our source. For details on the specific websites used for each state, refer to [Appendix C](#) and [Appendix D](#).

- The improved "document reference" and "document name" data means that those columns will be different from previous releases, but the document ID and metrics have largely remained the same. Be aware of these changes when attempting to merge the new release with older data. For any new projects, we suggest using this release in full, rather than merging with older data.
- Enhancements to document-level data have been implemented up to 2020, with work ongoing for the remaining years. Document-level joinable data for 2019 and earlier will be included in future releases, which will allow users to compare documents across time for every year for which we have data.
- Some iterative improvements were made to the historical data (i.e., pre-2023). Ensure that you use the data in this release for ongoing projects to maintain accuracy and currency. The following states had large improvements that especially should be incorporated into any going research: District of Columbia, Michigan, Missouri, New Hampshire.

5 Contact

If you have any questions concerning our documents, data, methodology, or anything else about State RegData 2023, please email us at info@quantgov.org.

6 Citation

If you use this data, please cite:

McLaughlin, Patrick A., Michael Gilbert, Jonathan Nelson, and Thurston Powers. State RegData 2023 (dataset). QuantGov, Mercatus Center at George Mason University, Arlington, VA, 2023.

Current Version: 5.0

A Metadata Columns

Column	Description
document reference	The reference in the code where the document exists.
document name	A description of what the document is about.
agency	The agency responsible for writing and enforcing the regulation or statute.
agency cluster ^a	Category of the agency.

^aSee section 2.1.1 above for more information about agency clusters.

B Data Series

Data Series	Description
words	Total number of words.
shall	Occurrences of the word shall.
must	Occurrences of the word must.
may not	Occurrences of the words may not.
required	Occurrences of the word required.
prohibited	Occurrences of the word prohibited.
restrictions	Total number of the five restrictive terms above.
sentence length	Average sentence length in the document.
shannon entropy	An estimation of amount of unique information within the document.
conditionals per 100 sentences	Number of conditionals ^a per 100 sentences.
long word score	Average squared word length for the document.
industry relevance	Probability that the document is relevant to a given industry.
industry relevant restrictions ^b	Number of restrictions relevant to a specific industry.

^aConditionals include the terms "if", "but", "except", "provided that", "when", "where", "whenever", "unless", "notwithstanding", "in the event", "in no event", and "to the extent that"

^bCalculated by multiplying "restrictions" by the document's "industry relevance" per industry.

C Websites Used: Regulations

State	Website
Alabama	https://casetext.com/library
Alaska	https://casetext.com/library
Arizona	https://casetext.com/library
Arkansas	N/A
California	https://govt.westlaw.com/calregs/Index
Colorado	https://casetext.com/library
Connecticut	https://casetext.com/library
Delaware	https://casetext.com/library
D.C.	https://casetext.com/library
Florida	https://www.flrules.org/gateway/browse.asp
Georgia	https://casetext.com/library
Hawaii	https://casetext.com/library
Idaho	https://adminrules.idaho.gov/rules/current/index.html
Illinois	https://www.ilga.gov/commission/jcar/admincode/titles.html
Indiana	http://iac.iga.in.gov/iac/iac_title
Iowa	https://www.legis.iowa.gov/law/administrativeRules/agencies
Kansas	https://casetext.com/library
Kentucky	https://casetext.com/library
Louisiana	https://casetext.com/library
Maine	https://casetext.com/library
Maryland	https://casetext.com/library
Massachusetts	https://casetext.com/library
Michigan	https://casetext.com/library
Minnesota	https://www.revisor.mn.gov/rules/
Mississippi	https://www.sos.ms.gov/adminsearch/default.aspx
Missouri	https://casetext.com/library
Montana	http://www.mtrules.org/gateway/browse.asp

State	Website
Nebraska	https://casetext.com/regulation/nebraska-administrative-code
Nevada	https://www.leg.state.nv.us/nac/CHAPTERS.HTML
New Hampshire	http://www.gencourt.state.nh.us/rules/about_rules/listagencies.aspx
New Jersey	https://casetext.com/library
New Mexico	https://casetext.com/library
New York	https://govt.westlaw.com/nycrr/
North Carolina	http://reports.oah.state.nc.us/ncac.asp
North Dakota	https://www.ndlegis.gov/agency-rules/north-dakota-administrative-code/
Ohio	http://codes.ohio.gov/oac/
Oklahoma	https://casetext.com/library
Oregon	https://casetext.com/library
Pennsylvania	https://casetext.com/library
Rhode Island	https://casetext.com/library
South Carolina	https://casetext.com/library
South Dakota	https://casetext.com/library
Tennessee	https://sos.tn.gov/publications/
Texas	https://casetext.com/library
Utah	https://casetext.com/library
Vermont	https://casetext.com/library
Virginia	https://casetext.com/library
Washington	https://app.leg.wa.gov/wac/
West Virginia	https://apps.sos.wv.gov/adlaw/csr/
Wisconsin	https://casetext.com/library
Wyoming	https://rules.wyo.gov/Search.aspx?mode=1

D Websites Used: Statutes

State	Website
Alabama	https://casetext.com/library
Alaska	https://casetext.com/library
Arizona	https://casetext.com/library
Arkansas	https://casetext.com/library
California	https://casetext.com/library
Colorado	https://casetext.com/library
Connecticut	https://casetext.com/library
Delaware	https://casetext.com/library
D.C.	https://casetext.com/library
Florida	https://casetext.com/library
Georgia	https://casetext.com/library
Hawaii	https://www.capitol.hawaii.gov/hrscurrent/
Idaho	https://casetext.com/library
Illinois	https://www.ilga.gov/legislation/ilcs/ilcs.asp
Indiana	https://casetext.com/library
Iowa	https://casetext.com/library
Kansas	https://casetext.com/library
Kentucky	https://casetext.com/library
Louisiana	https://casetext.com/library
Maine	https://casetext.com/library
Maryland	https://casetext.com/library
Massachusetts	https://casetext.com/library
Michigan	https://casetext.com/library
Minnesota	https://casetext.com/library
Mississippi	https://casetext.com/library
Missouri	https://casetext.com/library
Montana	https://leg.mt.gov/bills/mca/index.html

State	Website
Nebraska	https://nebraskalegislature.gov/laws/browse-statutes.php
Nevada	https://www.leg.state.nv.us/nrs/
New Hampshire	https://casetext.com/library
New Jersey	https://casetext.com/library
New Mexico	https://casetext.com/library
New York	https://casetext.com/library
North Carolina	https://casetext.com/library
North Dakota	https://casetext.com/library
Ohio	https://codes.ohio.gov/ohio-revised-code
Oklahoma	https://www.oscn.net/applications/oscn/index.asp?ftdb=STOKST
Oregon	https://casetext.com/library
Pennsylvania	https://casetext.com/library
Rhode Island	https://casetext.com/library
South Carolina	https://casetext.com/library
South Dakota	https://casetext.com/library
Tennessee	https://casetext.com/library
Texas	https://casetext.com/library
Utah	https://casetext.com/library
Vermont	https://casetext.com/library
Virginia	https://law.lis.virginia.gov/vacode/
Washington	https://apps.leg.wa.gov/rcw/
West Virginia	https://casetext.com/library
Wisconsin	https://docs.legis.wisconsin.gov/statutes/prefaces/toc
Wyoming	https://casetext.com/statute/wyoming-statutes